

Answers 3.6

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1.

The screenshot shows a PostgreSQL query editor interface. On the left, the 'Object Explorer' pane displays a tree structure of databases, with 'ANSWERS 3.2' selected under the 'PostgreSQL 17' server. The main editor area shows a SQL query:

```
1 SELECT title, release_year, language_id, rental_duration,  
2 COUNT(*)  
3 FROM film  
4 GROUP BY title, release_year, language_id, rental_duration  
5 HAVING COUNT (*) > 1;  
6
```

Below the query, the 'Data Output' pane shows the results of the query. The output is a table with the following columns:

title	release_year	language_id	rental_duration	count
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The status bar at the bottom indicates 'Total rows: 0', 'Query complete 00:00:00.262', and 'Ln 5, Co'.

How I would clean data if I had to-

I would use the DISTINCT to check for any duplication from the columns.

Use DISTINCT and GROUP BY as keywords to cover up for non_uniform data.

I would manually if I have access to the source of the data and use the UPDATE command.

I would use the impute approach for missing values using statistical methods.

2.

The screenshot shows a PostgreSQL client interface with the following components:

- Object Explorer:** A tree view on the left showing the database structure. Under 'PostgreSQL 17', the 'Databases (3)' folder is expanded, showing 'ANSWERS 3.2', 'Rockbuster', and 'postgres'. 'ANSWERS 3.2' is selected.
- Query Editor:** The main area for writing SQL queries. It contains the following SQL code:

```
1 SELECT
2 --rental_rate
3 MIN(rental_rate) AS min_rent,
4 MAX(rental_rate) AS max_rent,
5 AVG(rental_rate) AS avg_rent,
6 COUNT (rental_rate) AS count_rent_values
7 FROM film
```
- Execute script:** A button with a play icon and the text 'Execute script' (F5) is located above the query editor.
- Data Output:** The results of the query are displayed in a table. The table has four columns: 'min_rent' (numeric), 'max_rent' (numeric), 'avg_rent' (numeric), and 'count_rent_values' (bigint). The first row shows the results for the 'film' table.

	min_rent numeric	max_rent numeric	avg_rent numeric	count_rent_values bigint
1	0.99	4.99	2.9800000000000000	1000
- Footer:** At the bottom, it says 'Total rows: 1' and 'Query complete 00:00:00.108'.

rental rate

Object Explorer

Servers (4)

PostgreSQL 13

PostgreSQL 14

PostgreSQL 15

PostgreSQL 17

Databases (3)

ANSWERS 3.2

Rockbuster

postgres

Login/Group Roles

Tablespaces

pgAgent Jobs

encies × Dependents × Processes × SQL × Rockbuster/postgres@PostgreSQL 17* ×

Rockbuster/postgres@PostgreSQL 17

Query

Query History

1 SELECT

2 MIN(length) AS min_length,

3 MAX(length) AS max_length,

4 AVG(length) AS avg_length,

5 COUNT(length) AS count_length_values

6 FROM film

Data Output

Messages

Explain ×

Notifications

Showing rows: 1 to 1

Page No: 1

of 1

	min_length smallint	max_length smallint	avg_length numeric	count_length_values bigint
1	46	185	115.2720000000000000	1000

Total rows: 1

Query complete 00:00:00.211

LF

Ln 5, Col 30

length

Object Explorer

Servers (4)

PostgreSQL 13

PostgreSQL 14

PostgreSQL 15

PostgreSQL 17

Databases (3)

ANSWERS 3.2

Rockbuster

postgres

Login/Group Roles

Tablespaces

pgAgent Jobs

encies × Dependents × Processes × SQL × Rockbuster/postgres@PostgreSQL 17* ×

Rockbuster/postgres@PostgreSQL 17

Query

Query History

1 SELECT

2 MIN(release_year) AS min_release_year,

3 MAX(release_year) AS max_release_year,

4 AVG(release_year) AS avg_release_year,

5 COUNT(release_year) AS count_release_year_values

6 FROM film

Data Output

Messages

Explain ×

Notifications

Showing rows: 1 to 1

Page No: 1 of 1

	min_release_year integer	max_release_year integer	avg_release_year numeric	count_release_year_values bigint
1	2006	2006	2006.0000000000000000	1000

Total rows: 1

Query complete 00:00:00.250

LF

Ln 5, Col 42



release year

Object Explorer

Servers (4)

PostgreSQL 13

PostgreSQL 14

PostgreSQL 15

PostgreSQL 17

Databases (3)

ANSWERS 3.2

Rockbuster

postgres

Login/Group Roles

Tablespaces

pgAgent Jobs

encies × Dependents × Processes × SQL × Rockbuster/postgres@PostgreSQL 17* ×

Rockbuster/postgres@PostgreSQL 17

No limit

Execute script

F5

Query Query History

1 SELECT

2 MIN(rental_duration) AS min_rent_duration,

3 MAX(rental_duration) AS max_rent_duration,

4 AVG(rental_duration) AS avg_rent_duration,

5 COUNT(rental_duration) AS count_rent_duration_values

6 FROM film

Data Output Messages Explain × Notifications

Showing rows: 1 to 1

Page No: 1 of 1

	min_rent_duration smallint	max_rent_duration smallint	avg_rent_duration numeric	count_rent_duration_values bigint
1	3	7	4.9850000000000000	1000

Total rows: 1 Query complete 00:00:00.205 LF Ln 5, Col 22



rental duration

The screenshot shows a PostgreSQL client interface with the following components:

- Object Explorer:** A tree view on the left showing the database structure. It includes servers (PostgreSQL 13, 14, 15, 17) and databases (ANSWERS 3.2, Rockbuster, postgres). The 'ANSWERS 3.2' database is selected.
- Query Editor:** The main area for writing SQL queries. It contains the following query:


```

      1 SELECT
      2 MIN(replacement_cost) AS min_rep_cost,
      3 MAX(replacement_cost) AS max_rep_cost,
      4 AVG(replacement_cost) AS avg_rep_cost,
      5 COUNT(replacement_cost) AS count_rep_values
      6 FROM film
      
```
- Data Output:** A table showing the results of the query. It has 4 columns: min_rep_cost (numeric), max_rep_cost (numeric), avg_rep_cost (numeric), and count_rep_values (bigint). The results are:

	min_rep_cost numeric	max_rep_cost numeric	avg_rep_cost numeric	count_rep_values bigint
1	9.99	29.99	19.984000000000000000	1000
- Status Bar:** At the bottom, it shows 'Total rows: 1', 'Query complete 00:00:00.204', and 'LF Ln 6, Col 10'.

replacement cost

3.

I said in the previous text that I did find Excel much more easy fast and easy to work with but after diving a bit deeper into SQL, I'm having second thoughts about what I said earlier on because SQL is proving to be more effective when it comes to profiling by using the correct commands so with that being said, I find it to be much better than Excel.